

Study Notes: Introduction to Quantum Algorithms, Circuits, and Systems

1. Foundations of Quantum Circuits

The **Circuit Model** is the standard framework for quantum computation, defining a sequence of operations on **wires** (qubits) using **logic gates**. Unlike classical bits, quantum gates must be reversible, leading to the **Unitary Requirement**.

Mathematical Unitary Definition

Every quantum gate is represented by a unitary matrix U . A matrix is unitary if it satisfies: the conjugate transpose (adjoint) and is the identity matrix. This ensures the preservation of the L2 norm (total probability remains 1) and enables the reversibility of the computation.

Single Qubit Gates

Single qubit gates represent rotations on the **Bloch Sphere**.

Pauli X

Pauli Y

Pauli Z

Hadamard (H)

S Gate

Basis Shifting

The Hadamard gate is the primary tool for shifting between the **Computational (Z) Basis** ($|0\rangle, |1\rangle$) and the **Superposition (X) Basis**. Applying H allows an algorithm to process information in the X-basis, which is vital for interference-based algorithms like Deutsch-Jozsa.

2. Multi-Qubit Systems and Entanglement

Tensor Products

Composite systems are represented using the **Tensor Product**. For an n -qubit system, the state resides in a 2^n -dimensional Hilbert space.

Entanglement vs. Product States

- **Uncoordinated (Product) States:** States that can be factored into individual qubit states. Measurement of one qubit provides no information about the other.
- **Correlated (Entangled) States:** States that cannot be factored. These exhibit non-classical correlations where the state of the whole is more precisely defined than the state of its parts.

Quantum Teleportation Protocol

This protocol transfers an unknown state using a shared Bell pair and classical communication. Importantly, this adheres to the **No-Cloning Theorem**: the original state is destroyed to reappear at the destination, a process analogous to "The Prestige" (original state collapse).

- **Entanglement:** Alice and Bob share state. Alice holds a state.

- **Alice's Operation:** Alice performs a Bell Measurement on $|n\rangle$ and her half of the Bell pair.
- **Classical Transmission:** Alice sends two classical bits to Bob.
- **Bob's Correction:** Bob applies Pauli gates based on the bits received:
 - **00:** Apply Identity
 - **01:** Apply X gate.
 - **10:** Apply Z gate.
 - **11:** Apply Zx (or Y) gate.
- **Result:** Bob's qubit

3. Quantum Algorithms I: Deutsch-Jozsa

The Deutsch-Jozsa algorithm determines if a function is **Constant** (all outputs 0 or 1) or **Balanced** (exactly half 0, half 1).

Complexity and Phase Kickback

- **Classical Complexity:** Requires $2^{n-1} + 1$ queries for 100% certainty.
- **Quantum Complexity:** Requires only **1 query**.
- **Phase Kickback:** The algorithm utilizes an auxiliary qubit in the $|0\rangle$ state. When the oracle is applied, the auxiliary qubit remains unchanged, but its phase is "kicked back" to the main register.

Circuit Stages

1. **Initialization:** Prepare n qubits in $|0\rangle^{\otimes n}$ and an auxiliary qubit in $|-\rangle$ (via X then H).
2. **Superposition:** Apply $H^{\otimes n}$ to the register.
3. **Oracle:** Apply the phase oracle U_f .
4. **Measurement:** Apply H gate and measure. A result indicates a **Constant** function; any other result indicates a **Balanced** function.

4. Quantum Algorithms II: Grover's Search

Grover's algorithm performs an unstructured search in $O(N^{1/2})$ time, marking a quadratic speedup over classical $O(N)$ search.

Core Components

1. **Phase Oracle:** Flips the sign of the marked element $|w\rangle$ to $-|w\rangle$.
2. **Diffuser:** Performs inversion about the mean to amplify the marked element's probability amplitude.

Hardware Performance Insights

- **Simulator:** Achieves near-ideal results (e.g., **99.9%** probability of finding the target).
- **Actual Hardware (e.g., IBM Belem/Melbourne):** Noise significantly impacts depth. In lab trials, the probability of finding the marked element on real hardware can drop **below 50%**, making multiple **shots** (8192) essential to resolve the signal.

5. Variational Quantum Algorithms (VQE & QAOA)

Parameterized Quantum Circuits (PQC)

PQCs use Parameter objects to allow gate angles to be adjusted by a classical optimizer. This hybrid approach is the cornerstone of Near-Intermediate Scale Quantum (NISQ) applications.

QAOA Structure and Periodicity

QAOA solves combinatorial optimization by alternating two layers:

1. **Cost Layer** : Encodes the problem (e.g., Max-Cut).
2. **Mixer Layer** : Drives transitions between states.**Parameter Periodicity:**
 - **Mixer:** Periodicity is fixed due to the nature of RX gates.
 - **Cost:** Periodicity depends on problem weights. For integer-weighted Max-Cut, the energy landscape repeats.

Classical Optimizers

The variational loop utilizes optimizers to find the ground state:

- **COBYLA:** Gradient-free; robust against noise but prone to local optima.
- **Adam:** Gradient-based; effective for complex landscapes.
- **SLSQP:** Good for constrained optimization.

Max-Cut as QUBO

Max-Cut is a **Quadratic Unconstrained Binary Optimization (QUBO)** problem. The goal is to partition a graph to maximize the weight of edges between sets. This is mapped to a Hamiltonian using **ZZ-interactions**, where the quantum state with the lowest energy corresponds to the optimal cut.

6. Quantum Noise and Imperfections

SPAM and Readout Error Matrix

State Preparation and Measurement (SPAM) errors occur at the beginning and end of a circuit. Readout errors are characterized by the **Confusion Matrix**.

Noise Scaling

- **Coherent Noise:** Systematic errors (e.g., miscalibrated rotations). Impact scales **quadratically**
- **Incoherent Noise:** Stochastic/random decoherence from environmental interaction. Impact scales **linearly** .

Projection Noise

Statistical fluctuations from a finite number of measurement shots. Maximum noise occurs at $P=0.5$ (superposition), while $P=0,1$ (basis states) have zero projection noise.

Density Matrices

To describe "open" quantum systems and **ensembles of trajectories** , we use density matrices. This allows us to track the statistics of mixed states where the exact pure state trajectory is unknown due to stochastic noise.

7. Practical Implementation Notes (Qiskit & IBM Quantum)

Job Execution Workflow

1. **Backend Selection:** Use `least_busy()` to minimize wait times.
2. **Optimization:** Setting `optimization_level=3` is recommended for hardware. This level performs aggressive gate cancellation and optimal qubit mapping.
3. **Job ID:** Vital for retrieving data after a session.

Hardware Constraints: Connectivity and SWAPs

The **Connectivity Map** is a critical hardware constraint. If an algorithm requires a gate between non-adjacent qubits, the compiler must insert **SWAP gates**. Because SWAPs are composed of three noisy CNOTs, they significantly increase circuit depth and decoherence. Selecting high-connectivity qubits from the map is essential to avoid these "noisy SWAP gates" and maintain fidelity.